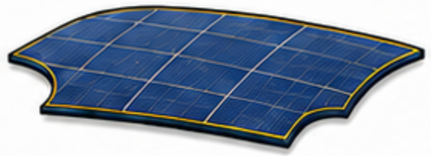
ILLUSTRATIVE SUIT DESIGN DETAIL

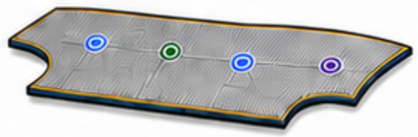
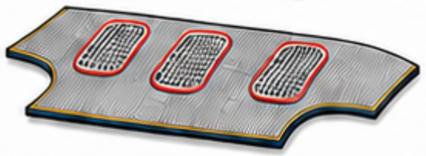
ENERGY LAYER

1. photovoltaic textile shell
2. isolation + charge regulation
3. removable storage

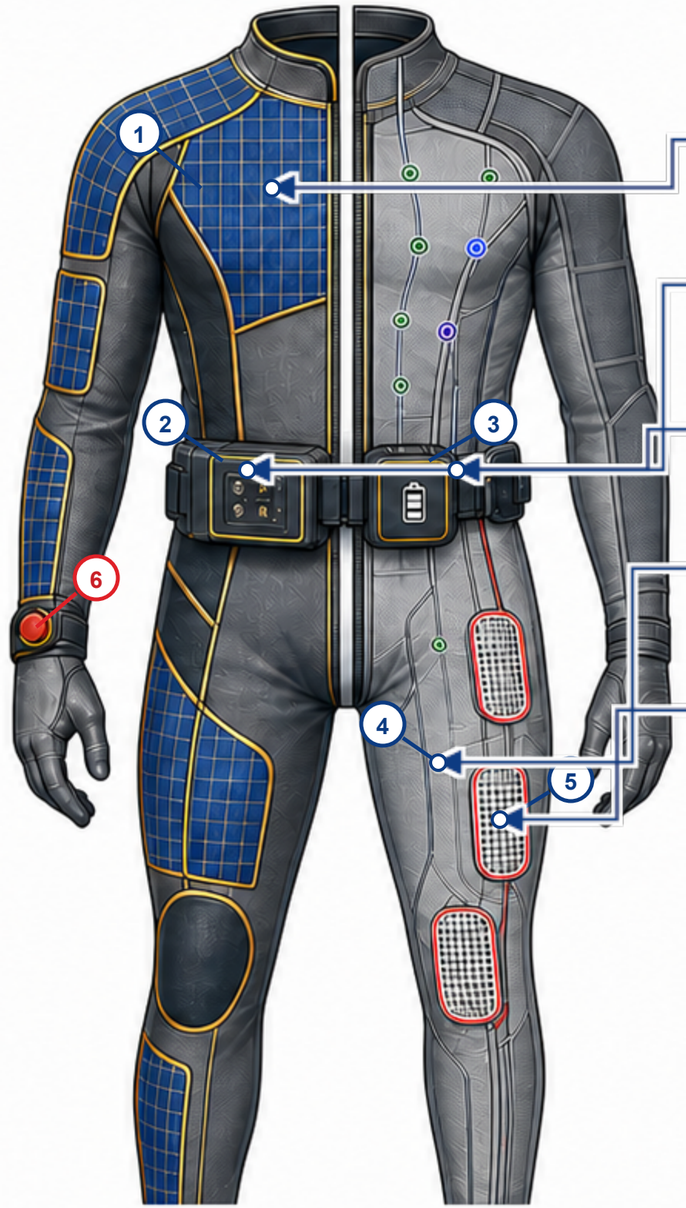
BODILY LAYER

4. contact / impedance sensing
5. soft NMES electrodes
6. local emergency stop




- illumination (energy in)
- reserve / power flow
- electrode contact / stimulation
- user command / decision
- physical stop path (independent)



PV OUTER SHELL
flexible light harvesting

CONTROLLER
isolation + charge regulation + NMES driver

REMOVABLE STORAGE
sealed removable battery

SENSOR NODES
contact + impedance temperature + motion

NMES ELECTRODES
soft pads over selected muscle groups

CONTROLLER / INTERFACE

- preview — user reviews system status, energy reserve, sensor health, and environment
- affirmative initiation — user affirms to arm NMES support for contingency use
- monitor — real-time monitoring of user state, energy reserve, and system performance
- stop / degrade — user or system stops or degrades NMES support as needed

USER + CLINICAL LIMITS

- not flight qualified
- no autonomous background stimulation
- contingency support only

PHYSICAL STOP PATH (INDEPENDENT OF DISPLAY)

- press local emergency stop (large tactile button)
- immediate hardware interrupt to NMES driver outputs
- all stimulation outputs disabled (regardless of controller state)

DATA + ENERGY FLOWS (LOGICAL)








illumination (energy in) → photovoltaic textile shell → isolation + charge regulation → removable storage → controller / stimulator (decision logic) → NMES electrodes (outputs)

sensing inputs (contact, impedance, temp., motion, env.) → controller / stimulator (decision logic)